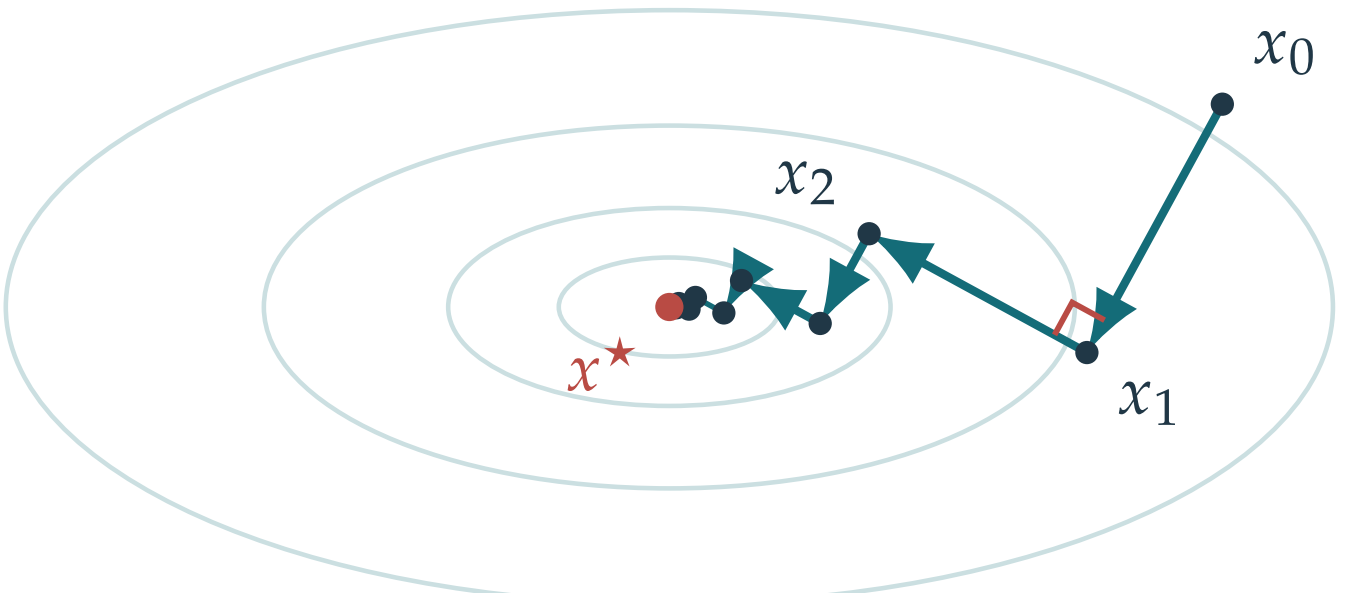
$$\tau_k \in \operatorname*{argmin}_{\tau \geq 0} f(x_k - \tau \nabla f(x_k))$$



$$\langle \nabla f(x_k), \nabla f(x_{k+1}) \rangle = 0$$

$$f(x) = \tfrac{1}{2}(x_1^2 + 5x_2^2)$$